

Patrick Serrano

Senior Backend Engineer (Golang / Data Infrastructure)

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Summary

Senior Backend Engineer with 7+ years of experience designing and operating high-throughput, low-latency financial systems and multi-tenant SaaS platforms. Expert in **Golang** and functional programming (Clojure), with a strong focus on event-driven architectures, distributed messaging (NATS JetStream), and bitemporal data infrastructure. Proven track record in building ultra-low latency engines ($\sim 43\mu s$) and managing complex financial data pipelines with millions of daily events. Specialized in "dual-store" architectural patterns, ensuring strict transactional consistency alongside robust analytical processing.

Skills

Languages	Golang (Concurrency, Low-Latency), Clojure, TypeScript, Python, SQL
Backend / Data	NATS JetStream, Redis Streams, XTDB (Bitemporal), Datomic (Datalog), PostgreSQL, DynamoDB, Onyx, Clara Rules
Architecture	High-Performance Computing, Event-Driven Architecture, Multi-tenant SaaS, CQRS, Polyolith, ETL Pipelines, Exactly-once Processing
Infra / DevOps	AWS (SQS, Lambda, ECS, SSM), Docker, Pulumi (IaC), CI/CD, Github Actions, Observability
Financial Tech	Payment Gateways (Adyen, Cielo, eRede), Anti-Fraud (ClearSale, Konduto), Automated Trading Engines, Transactional Auditing

Professional Experience

Jul 2024 – Present **Lead Architect & Developer (Ark Engine)**, *Quantitative Trading Infrastructure*

High-performance automated trading engine focused on ultra-low latency, bitemporal auditability, and absolute financial data resilience.

Highlights:

- **High-Performance Go Migration:** Redesigned the core decision engine from Clojure/XTDB to Golang/NATS JetStream to achieve extreme latency benchmarks, resulting in an end-to-end event chaining latency of $\sim 43\mu s$;
- **Concurrency & Distributed Messaging:** Leveraged Go's concurrency model (goroutines, channels) and NATS JetStream to build a fan-out topology with immutable snapshots, eliminating race conditions in high-frequency trading;
- **Bitemporal Persistence:** Implemented a "dual-store" pattern using XTDB to separate Valid Time from Transaction Time, structurally eliminating lookahead bias in backtesting and ensuring 100% auditability of financial decisions;
- **Exactly-Once Processing:** Designed event-driven pipelines with Redis Streams and NATS to guarantee atomic processing of market events across distributed consumer groups.

Technologies: Golang, NATS JetStream, Clojure, XTDB, Redis Streams, Docker, Event-Driven Architecture

2022 – 2024 **Senior Software Engineer (Octopus Pay)**, *Multi-Tenant Payment SaaS*

High-availability financial orchestration platform managing 25+ production integrations for e-commerce and payment gateways.

Highlights:

- **Data Ingestion Engine:** Architected "Summon," a high-throughput ETL engine for ingesting, enriching, and processing millions of financial events daily with strict delivery guarantees and circuit-breaker patterns;
- **Dual-Store Pattern:** Implemented a transactional-analytical architecture using Datomic (immutable log) and Pathom, allowing real-time derivation of hundreds of computed attributes via EQL/Datalog;
- **Adapter Architecture:** Designed and maintained 25+ canonical adapters for major payment gateways (Adyen, Cielo, Pagar.me) and anti-fraud systems, ensuring 99.99% uptime for critical financial flows;
- **Infrastructure as Code:** Managed cloud resources via Pulumi (AWS) and automated deployment pipelines, ensuring consistent environments for high-stakes payment processing;
- **Financial Logic:** Integrated forward-chaining rule engines (Clara Rules) to automate complex financial workflows and attribute derivation directly from transactional records.

Technologies: Clojure, Datomic, Pathom 3, Clara Rules, AWS, Pulumi, Reitit, Pedestal, Malli, Onyx

- 2020 – 2021

Backend Engineer (Zougue / MPMS), B2B Marketplace Platform

Enterprise-grade marketplace management platform integrated with major e-commerce players.

Highlights:

- **Integrations:** Built complex integration endpoints with VTEX for inventory, price, and order management, handling high-volume SKU updates and transactional synchronization;
 - **Data Modeling:** Designed multi-tenant Datomic schemas and Pathom resolvers to support multi-seller marketplace queries with real-time computed attributes;
 - **Automation:** Authored complex business rules for order routing and financial attribute computation using forward-chaining logic;
 - **Analytics:** Integrated Metabase for per-client analytical reporting, powered by Datalog queries against immutable data history.

Technologies: Clojure, Datomic, Pathom, Clara Rules, Malli, VTEX, Pedestal, Metabase
- 2018 – 2019

Software Engineer (Real Estate & Fintech), Startup Projects

- Developed full-stack applications from scratch, focusing on data consistency and secure communication flows for financial and real estate platforms;
 - Built initial integrations with financial systems, establishing a foundation in payment flow traceability and security best practices.

Technologies: Clojure, ClojureScript, Fulcro, React Native, REST APIs

Education

- 2024 – Present

Bachelor's in Software Engineering, Descomplica Faculdade Digital
- 2015 – 2018

Bachelor's in Physics, Universidade Federal Fluminense

Languages

- Portuguese

Native
- English

Advanced (Full Professional Fluency)
- German

Basic
- Spanish

Basic